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On the current conduction mechanisms of WO_3/n-Ge Schottky interfaces

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ABSTRACT

Electrical properties of *n*-Ge contacts tailored with nanostructured WO_3 is analysed using current–voltage (I-V) and capacitive-voltage (C-V) measurements. Au/WO_3/*n*-Ge heterostructure show its significant improvement in terms of high forward current and high magnitude of rectification ratio (RR = 3040 at ± 0.5V) than compared to contact without interlayer. The series resistance evaluated using Cheung's method found to be very low (R_s = 17 Ω) in contacts with WO_3 interlayer than compared to pristine sample. The influence of WO_3 interfacial layer in defining the accumulation and inversion regions is clearly observed in the C-V characteristics measured at 1 MHz. A transition from ohmic conduction at low bias voltages to trap charge limited current (TCLC) conduction at high bias voltages is observed in the case of Au/WO_3/*n*-Ge contacts. The distribution of traps or defects at the oxide-Ge interface possibly associated to the observed current conduction mechanisms at high bias regions. Very low series resistance, high rectification ratio and high forward currents of Au/WO_3/*n*-Ge contacts show its applicability in optoelectronic and photovoltaic applications.

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1. Introduction

Since the Silicon based CMOS technology is facing strong difficulty with the trends in enhancing the performance of the devices by increasing the CMOS scaling. Hence alternative technologies need to be identified to meet these technological demands for high speed device applications. Recently, Germanium based semiconductor devices received great attention due to its superior electron and hole mobility than its counterpart silicon based devices. These devices seems to be a good alternative to silicon based devices in view of designing high mobility CMOS devices. Metal-semiconductor contacts are proven an important basic element in the fabrication of various optoelectronic devices. The interfacial layer (IL) such as organic, metal-oxide materials in metal/IL/semiconductor contacts modifies the surface states of the semiconductor which further influence the device performance in terms of current transport properties. Schottky heterostructures with interfacial layers such as MIS or MOS devices are most attractive semiconductor devices in view of their enhanced electrical, optical, structural and interfacial properties. The interfacial layer considerably enhances the device properties in a significant way suitable for different applications like OLEDs, photodetectors, OFETs, gas sensors and photovoltaics [1–3]. In organic electronic devices, metal-oxide thin films are suitably used as an interfacial layer to improve the charge injection properties between semiconductor and metal [1]. In the absence of the oxide layer, metal and semiconductor experiences huge barriers for charge transport processes. Among various metal-oxides, few are characterized as hole transport materials (e.g., MoO_3, WO_3, V_2O_5, NiO, CuO) [4–6] and few as electron transport layers (e.g., ZnO, ZrO_2, TiO_2). Several reports preferably used thermal deposition methods under vacuum to fabricate the semiconductor devices using these metal-oxides which are lack in low cost and large area production of devices [5–8].

Among the various oxide materials, WO_3 shows its significance due to its higher bandgap, high transparency, high electron affinity, superior charge transfer characteristics and good absorber properties makes it suitable for optoelectronic applications [9]. p-Si based MIS contacts fabricated using Cerium doped WO_3 as interfacial layer using simple and inexpensive Jet Nebulisation Spray Pyrolysis Technique (JNSP) and studied its photodiode characteristics

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under different illuminations. They measured the electrical properties of these MIS contacts under illumination for different cerium concentrations [10]. Pd/WO_3/*n*-Si and Pd/WO_3/ZnO/*n*-Si heterojunctions were fabricated to understand the H_2 gas sensing properties of MIS contacts. They observed a reduction in barrier height of Pd/WO_3/*n*-Si contacts due to creation of dipole charges at the oxide and semiconductor interface significantly influence on the sensing properties [11]. Three TMOs such as MoO_3, V_2O_5, WO_3 were deposited using thermal evaporation on *n*-Si to fabricate MIS contacts and studied its electrical properties in a wide range of temperatures [12].

Recently, various reports are available on the fabrication of WO_3 based heterostructures like WO_3/p-typegraphene [13] for photocatalytic applications, WO_3/Ga_2S_3 heterojunctions fabricated on Yb substrates used as tunnel diodes and microwave resonators [14], WO_3/CuO heterostructure nanochannel fabricated using lithographic approaches for chemical sensor applications [15], highly sensitive ammonia sensor is fabricated using p-MoS_2/n-WO_3 heterojunctions [16]. On the other hand, Fermi level pinning in the metal/*n*-Ge Schottky interface is a serious problem in the fabrication of *n*-Ge based semiconductor devices. Reports are available in the fabrication of *n*-Ge based heterostructures which alleviate Fermi level depinning and improves the quality of the rectification behavior of Schottky contacts [8,17–20]. Khurelbaatar et al [17] fabricated Au/*n*-Ge contacts modified with monolayer Graphene as an interlayer and studies its temperature dependent electric measurements and found that the magnitude of barrier inhomogeneity is reduced in contacts with grapheme interlayer than Au/*n*-Ge contacts. This could be possibly associated with passivation of *n*-Ge surface using Graphene monolayer which reduces the Fermi level pinning of Au/*n*-Ge interface. Son et al [18] have studied the current transport processes of MoS_2/p-Ge hybrid devices in a wide temperature range. Liu et al [19] fabricated Ti/*n*-Ge Schottky contacts with graphene and fluorinated graphene interlayer and found that the contact with fluorinated graphene interlayer shows low barrier height than Ti/*n*-Ge contacts which could be possibly associated with depinning of Fermi level at Ti/*n*-Ge interface.

In the present report, an attempt is made to fabricate WO_3/*n*-Ge heterostructure with nanostructured WO_3 thin film on *n*-Ge using low-cost spin coating method. Electrical properties of the contacts were analysed using I-V and C-V measurements. The electrical properties of the contacts address the variation of Schottky barrier parameters such as barrier height, ideality factor, series resistance and also various current transport processes involved at WO_3-Ge interface in the forward bias voltages. A detailed analysis of electrical properties is very essential to understand the performance of the WO_3/*n*-Ge heterostructures and to identify its suitability to sensor, optoelectronic and photovoltaic applications.

2. Experimental methods

n-type Ge (1 0 0) (Sb-doped) wafers is utilized in this study with carrier concentration of 2×10^{18} cm^{-3}. Initially samples processed with trichloroethylene, acetone and methanol using ultrasonic agitator for 5 min. each followed by drying in nitrogen gas. Native oxide is removed from the semiconductor surface by degreasing the samples in H_2O_2:HF (100:1). Immediately, the wafer placed in the vacuum chamber to coat ohmic contact on the rough side of the sample with aluminium (Al) of 50 nm thickness. After ohmic contact formation, the samples were transferred to spin coater unit to coat WO_3 films on the smooth side of the *n*-Ge wafer. Nanoparticle tungsten oxide (WO_3) ink (product no. 793353) was purchased from sigma Aldrich. Spin coater is operated at 4000 rpm with 60 *sec* duration to coat the Nanoparticle ink on the polished side of the *n*-Ge surface. Gold (Au) with the thickness of 40 mins utilized as a Schottky contact and deposited on the top of WO_3 using shadow metallic mask [20]. A reference sample without interlayer is also fabricated with the same experimental procedure. I-V and C-V measurements of the contacts with and without WO_3 layer is measured in dark at room temperature.

3. Results and discussion

Current conduction properties of contacts with and without WO_3 interfacial layer are evaluated using I-V measurements performed at room temperature is shown in Fig. 1. Schottky structures with and without WO_3 interlayer shows excellent rectification characteristics. The forward current of the Au/WO_3/*n*-Ge heterostructures shows significant improvement than compared to Au/*n*-Ge Schottky contact. Assuming thermionic emission model, the barrier parameters were evaluated for the voltages ($V > 3kT/q$) in the linear segment of semilog I -V graph from slope and intercept. The barrier height and ideality factor of Au/*n*-Ge Schottky contact is 0.63 eV and 1.51 respectively and whereas for the contacts with WO_3 layer shows 0.59 eV and 1.1 respectively. The ideality factor close to unity for Au/WO_3/*n*-Ge heterostructure shows the current transport in the contacts is primarily controlled by diffusion mechanism [4]. The reduction in the barrier height and slightly enhanced reverse currents of Au/WO_3/*n*-Ge Schottky contact can be explained by the presence of thin WO_3 interlayer which creates fixed positive surface charge and surface states at the oxide–semiconductor interface [21].

Fig. 2 shows the rectification ratio of the Schottky contacts which clearly evidenced the significant improvement in the rectification ratio of contacts with WO_3 interlayer. The rectification ratio of the contacts at ± 0.5 V shows an increase from 42 for Au/*n*-Ge contacts to 3040 for Au/WO_3/*n*-Ge contacts. Remarkably, in the case of contacts with WO_3 interlayer, a high rectification ratio of 1×10^4 is achieved for ± 2 V. The increase in rectification ratio in the case of contacts with WO_3 interlayer signifies its importance in fabricating high quality *n*-Ge Schottky contacts. The observed rectification ratio of the contacts with WO_3 interlayer found to be superior than compared to the earlier reports published recently [22]. The reduction in the barrier height and increase in the forward current observed in WO_3/*n*-Ge heterojunctions found to be similar with earlier reports might be explained based on Fermi level depinning, creation of dipole charges at the WO_3-Ge interface and decreased tunneling resistance [8,23,19].

A nonlinear behavior observed at higher forward bias voltages caused to deviate the current conduction mechanism from ther-

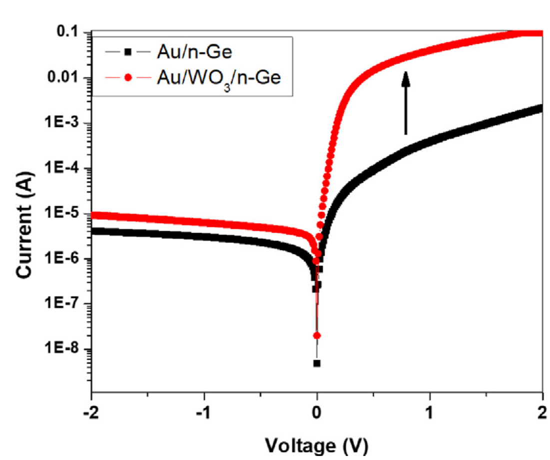

Fig. 1. I-V measurements of Au/Ge and Au/WO_3/*n*-Ge contacts.

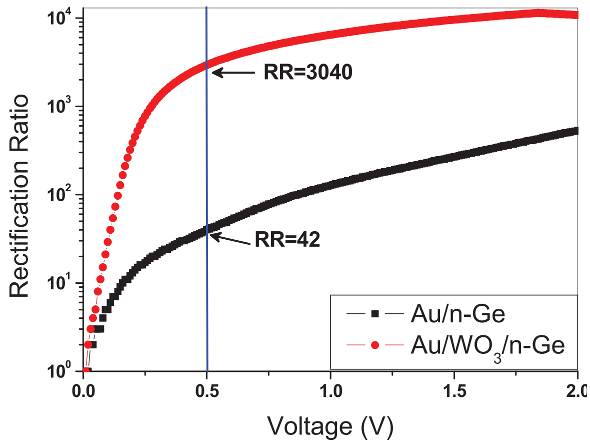

Fig. 2. Rectification Ratio of the contacts for constant bias voltage.

mionic emission. Due to this deviation, the barrier parameters might be quantified precisely by means of Cheung's method [24]. Fig. 3(a) and (b) shows the plot of dV/dlnI vs I and Cheung's function H(I) vs I graph for the pristine and WO_3/*n*-Ge contacts. The ideality factor and series resistance assessed from dV/dlnI vs I graph for Au/*n*-Ge sample is 1.97 and 3027 Ω respectively. For the contact with WO_3 layer, the series resistance was found to be drastically reduced to 17 Ω with ideality factor of 1.58. The decreased in series resistance observed in Au/WO_3/*n*-Ge sample may be the reason to increase the forward current to higher orders of magnitude. The Cheung's function (H(I)) vs I graph estimated the series resistance (R_s) of 2947 Ω and barrier height (Φ_b) of 0.61 eV for the Au/*n*-Ge samples. Similarly, a series resistance of 17 Ω and barrier height of 0.54 eV obtained for heterostructure sample with WO_3 interlayer.

The non-ideal current–voltage behavior of the Schottky contacts under different bias voltages shows linear regions at low biases and downward curvature region at higher biases. The difference in ideality factor observed in the different regions of the bias voltages could be ascribed to the bias dependent voltage drop across the WO_3 interlayer, surface states and series resistance. Norde [25] formulated a relation to estimate precisely the series resistance and barrier height for the Schottky contacts. Fig. 4 shows the Norde plot for the contacts with and without WO_3 interlayer. The evaluated Φ_b and R_s values are found to be 0.63 eV and 2458 Ω respectively for the Au/*n*-Ge Schottky contacts and 0.59 eV and 18 Ω respectively for WO_3/*n*-Ge Schottky structures. The details of the Schottky barrier parameters obtained from various methods are in full agreement and show its consistency with respect to thermionic emission model at low bias voltages. The details of the barrier parameters of pristine and heterostructure sample are mentioned in the Table 1.

The current conduction mechanism of the contacts during forward bias characteristics was investigated using logarithmic I-V plot as shown in Fig. 5. The linear relation between log I versus log V ($I \alpha V^m$) plot at different forward bias voltages shows different slope factors 'm' which quantifies the various conduction mechanisms. From the figure, the Au/*n*-Ge Schottky contact shows linear relation between voltage and current (with slope factor m = 1.63) which confirms ohmic conduction even for larger magnitudes of voltage. Two distinct regions were observed in case of contacts with WO_3 interlayer at low and moderately high voltages. The current conduction mechanism in heterojunction shows ohmic conduction (m = 1.4) at low voltages and a transition to trap charge limited current conduction (TCLC) (m = 3.56) is observed at moderately high voltages. This transition in conduction mechanism notably influences the current transport process across the heterojunction. A high forward current observed in Au/*n*-Ge contacts with WO_3 interlayer may be due to the presence of traps distributed exponentially at the oxide–semiconductor interface [26–28].

The C-V graph of the Schottky contacts with and without WO_3 interlayer measured at 1 MHz is shown in Fig. 6. As shown in figure, WO_3 interlayer preferably modifies the junction capacitance and it is clearly observed the variation in accumulation and inversion regions in the C-V characteristics. A sharp transition of capacitance is observed in WO_3/*n*-Ge heterostructures for the bias voltages changed from large forward voltages to very small reverse voltages. Further very low junction capacitance observed in contacts with WO_3 interlayer in the accumulation region could be due to the reduction in the density of states at oxide-Ge interface [29]. The negative capacitance observed in these samples may be

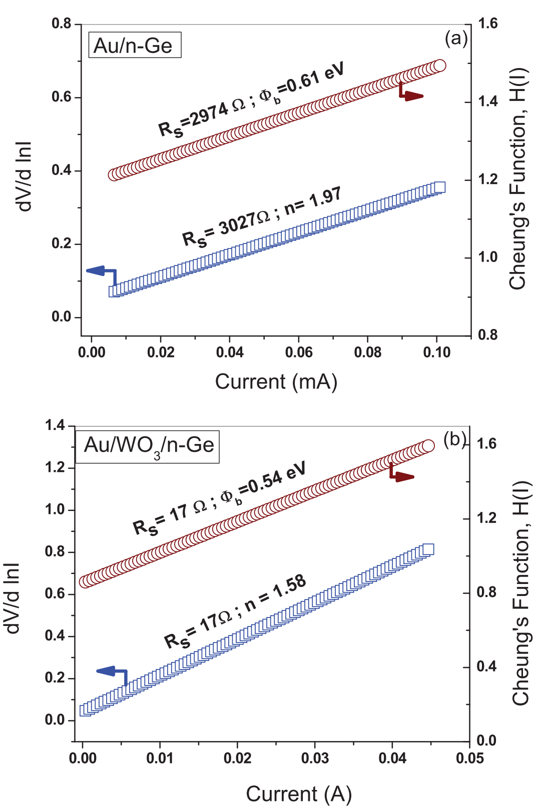

Fig. 3. (a) dV/dlnI vs I plot and (d) Cheung's function H(I) vs I plot.

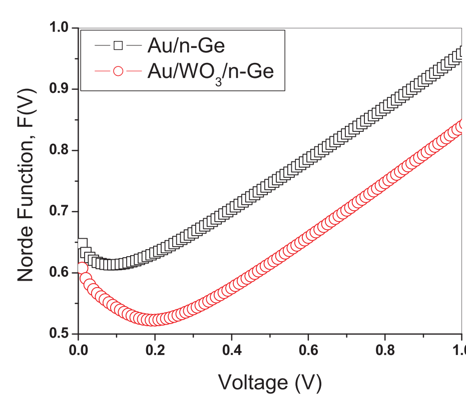

Fig. 4. Norde plot of Au/*n*-Ge and Au/WO_3/*n*-Ge Schottky contacts.

Table 1
Summary of Schottky contact parameters obtained from different methods.

Sample/ Method/ Parameters	Semilog I-V		dV/d lnI vs I		H(I) vs I		Norde Plot	
	n	Φ_b (eV)	R_S (Ω)	n	R_S (Ω)	Φ_b (eV)	R_S (Ω)	Φ_b (eV)
Au/*n*-Ge	1.51	0.63	3027 Ω	1.97	2974 Ω	0.61	2458 Ω	0.63
Au/WO_3/*n*-Ge	1.1	0.59	17 Ω	1.58	17 Ω	0.54	18 Ω	0.59

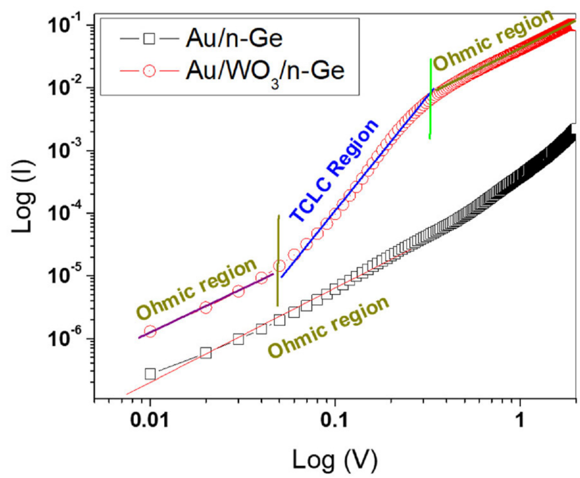

Fig. 5. Current conduction properties of contacts.

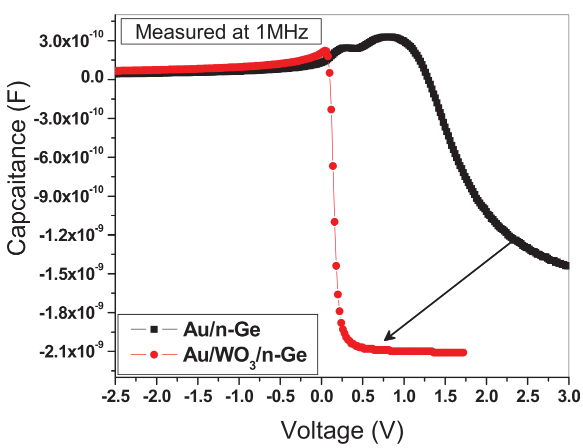

Fig. 6. Capacitance-Voltage measurements of contacts.

associated with the injection of minority carriers at the contact and interface states [30].

The observed variations of enhanced forward current, low barrier height in the case of contacts with WO_3 interlayer can be explained based on the creation a dipole layer with strong charge inversion at the WO_3 and *n*-Ge interface [31]. This dipole layer in turn changes the band offset between oxide–semiconductor interface causes to increase the current conduction across the junction and hence improves the rectification characteristics of the contact with WO_3 interlayer than pristine sample [32–33]. Such dipoles increase semiconductor Fermi level and significantly reduce the barrier height by pinning the valence band edge of the semiconductor. This possibly improves the electrons migration from semiconductor to oxide layer through traps [8]. The most recent report makes it clear that the possibility of the traps at the oxide–semiconductor junction might be associated with the oxygen vacancies observed in the oxide layer [5].

4. Conclusion

Au/*n*-Ge heterostructures are realized with nanostructured WO_3 thin film using low-cost spin coating method. Electrical properties of the contacts with WO_3 interlayer shows improvement in forward current, low series resistance and large rectification ratio than compared to Au/*n*-Ge contact. Following are the importance findings of the proposed WO_3/*n*-Ge heterostructures:

- The barrier height and ideality factor of Au/*n*-Ge contact found to be 0.63 eV and 1.51 and 0.59 eV and 1.1 for Au/WO_3/*n*-Ge heterostructure respectively.
- The consistency of the barrier parameters was also checked by Cheung and Norde methods and found low series resistance for Au/WO_3/*n*-Ge contacts than Au/*n*-Ge contacts.
- The observed variation of electrical properties of the contacts with WO_3 interlayer could be associated with the alleviation of Fermi level pinning at the *n*-Ge surface and reduction in the density of states at the oxide-Ge interface.
- The C-V measurements performed at 1 MHz shows the variation of accumulation and inversion zones for the heterostructure sample demonstrates the influence of WO_3 layer between metal and semiconductor.
- The current transport processes of the contacts with and without WO_3 interlayer is evaluated using logarithmic I-V plot in the forward bias region.
- A transition in the current transport process from ohmic to TCLC is observed in the case of the Au/WO_3/*n*-Ge heterostructures could possibly attributed to the exponential distribution of traps or defects at the oxide-Ge interface.
- This experimental result shows that the WO_3 interlayer influences significantly the transport properties and enhances the performance of the Au/WO_3/*n*-Ge heterostructures shows its applicability in photovoltaic and optoelectronic applications.

CRediT authorship contribution statement

G. Henry Thomas: Methodology, Data curation, Formal analysis, Writing – original draft. **A. Ashok Kumar:** Methodology, Data curation, Validation, Writing – review & editing. **V. Rajagopal Reddy:** Methodology, Validation, Writing – review & editing. **V. Janardhanam:** Resources, Data curation, Validation. **Chel-Jong Choi:** Resources, Data curation, Validation.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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